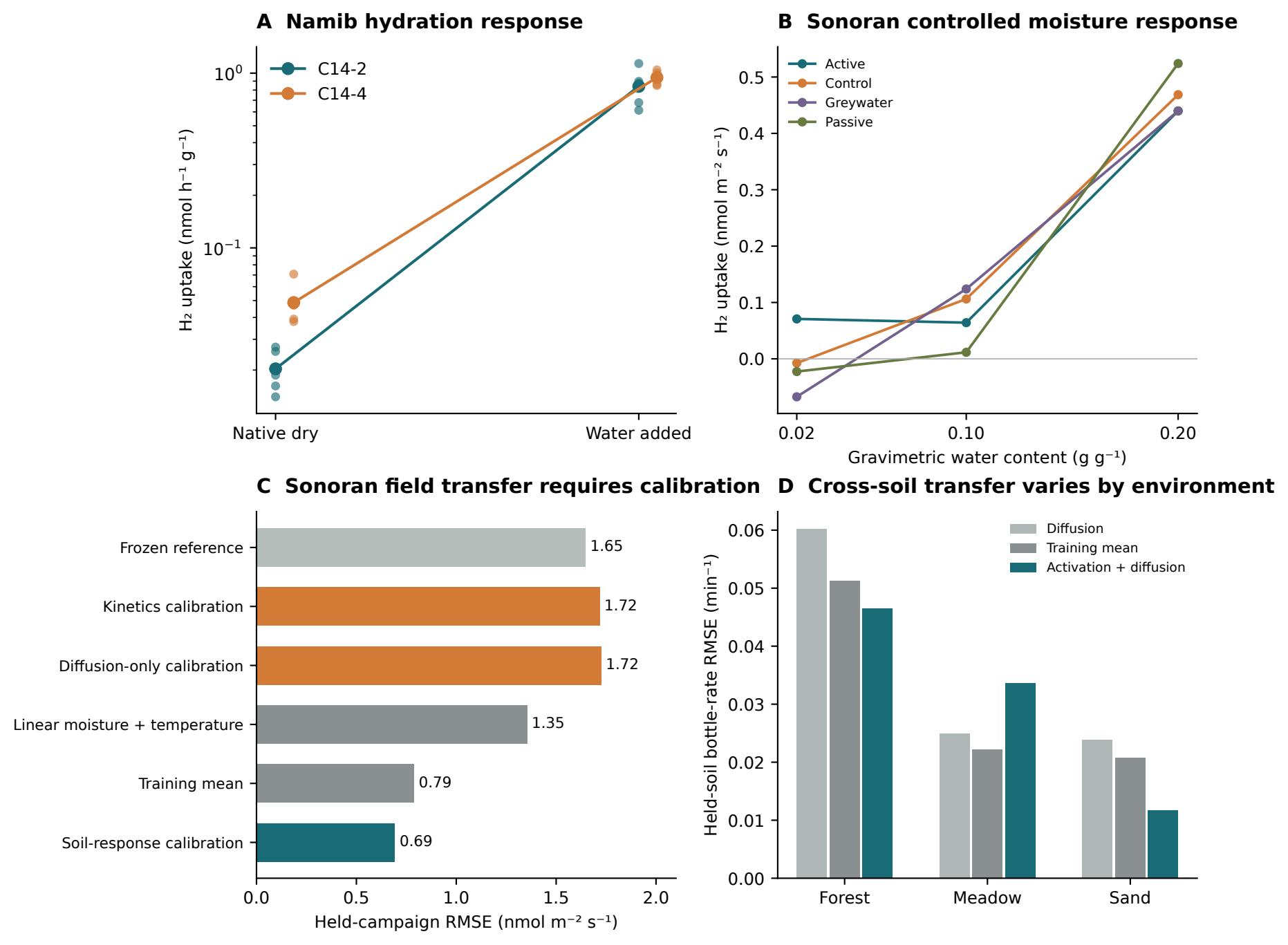
A Namib hydration response
C14-2
C14-4
10^0
10^{-1}
H₂ uptake (nmol h⁻¹ g⁻¹)
Native dry
Water added
B Sonoran controlled moisture response
Active
Control
Greywater
Passive
H₂ uptake (nmol m⁻² s⁻¹)
0.5
0.4
0.3
0.2
0.1
0.0
0.02
0.10
0.20
Gravimetric water content (g g⁻¹)
C Sonoran field transfer requires calibration
Frozen reference
Kinetics calibration
Diffusion-only calibration
Linear moisture + temperature
Training mean
Soil-response calibration
1.65
1.72
1.72
1.35
0.79
0.69
0.0
0.5
1.0
1.5
2.0
Held-campaign RMSE (nmol m⁻² s⁻¹)
D Cross-soil transfer varies by environment
Diffusion
Training mean
Activation + diffusion
Held-soil bottle-rate RMSE (min⁻¹)
0.06
0.05
0.04
0.03
0.02
0.01
0.00
Forest
Meadow
Sand